

# PARIKSHAK

An on-board witness that checks every step of an experiment procedure in orbit — and says so when it cannot see.

|                         |                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------|
| Problem Statement ID    | 26174                                                                                                              |
| Problem Statement Title | AI-based Human Activity Recognition for validating scientific experiment sequences on-board (BAS / lunar missions) |
| Organisation            | Indian Space Research Organisation (ISRO), Department of Space                                                     |
| Theme                   | Space Technology                                                                                                   |
| PS Category             | Software                                                                                                           |
| Team ID                 | [fill in your team ID]                                                                                             |
| Team Name               | [fill in your team name]                                                                                           |

# Proposed solution

PARIKSHAK watches the experiment rack through a camera, checks every step against the procedure file, speaks the next step, and alerts only on strong evidence — naming the step and the reason.

## How it addresses the problem

- Verifies each step independently — the crew no longer self-reports on a tablet.
- Catches skipped steps, the wrong object and safety violations within about 2 s.
- Says “cannot verify” when blind: an occluded camera never makes an accusation.
- Writes a tamper-evident flight record that is downlinked before any video.

## Innovation and uniqueness

### The experiment is a file

A new procedure is a YAML file. CRX-2 was added with no retraining and no code change.

### Unknown is not wrong

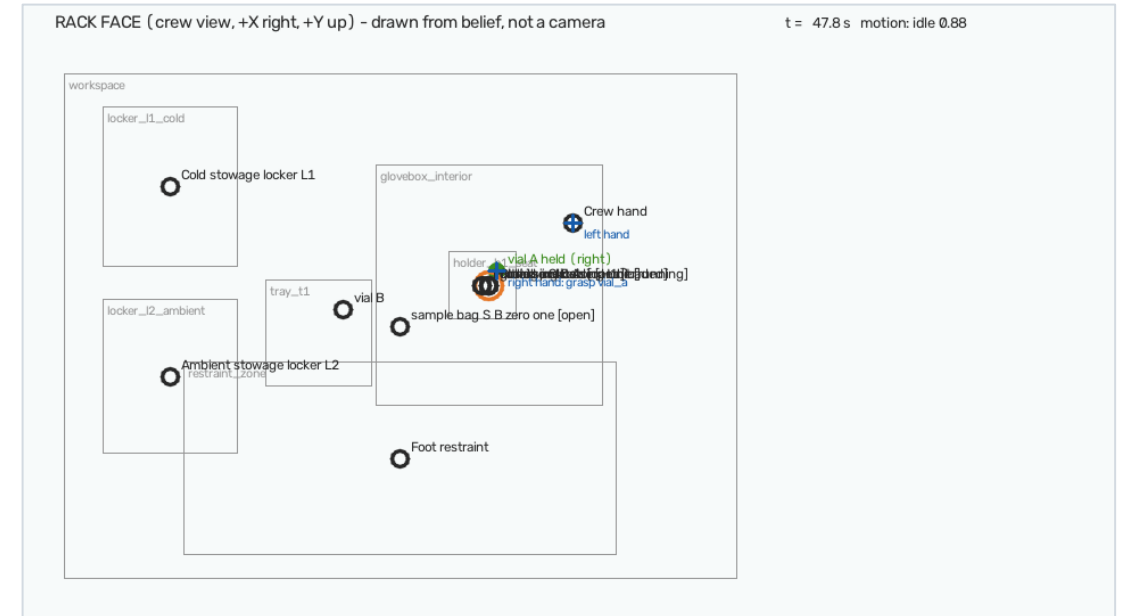
Three-valued logic: hidden or unmodelled evidence reads “cannot verify”, never “skipped”.

### Up is the rack, not gravity

Geometry is measured in rack coordinates from markers, so orientation in orbit does not matter.

### Evidence over time

Each step accumulates evidence with duration priors; an alert needs 0.85 evidence held for 1.5 s.

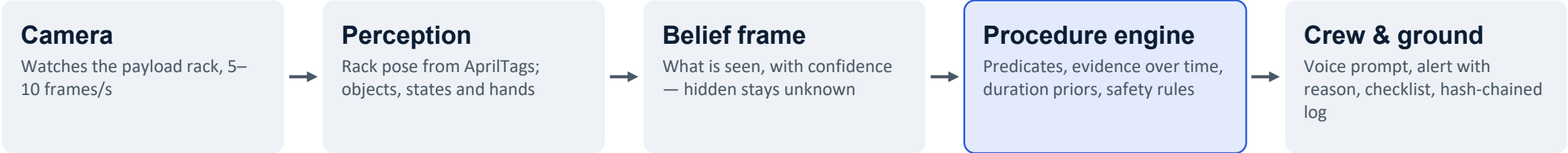


**CAUTION · SKIP · S08 · 47.8 s**

**“Step eight, close the glovebox latch, was not completed.”**

**Why:** predicate `state_is(latch, closed)` never satisfied between 43.0s and 46.2s

# Technical approach



| Layer      | Working prototype                                                                          | Flight version (planned)                                  |
|------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Perception | OpenCV 5, AprilTag 36h11 markers, PnP rack pose                                            | RT-DETR objects, MediaPipe / RTMPose hands, TensorRT INT8 |
| Reasoning  | Procedure language (YAML + JSON Schema), 3-valued predicates, CUSUM evidence, alert policy | Same engine, unchanged                                    |
| Interface  | Guided web app (FastAPI), PySide6 window, pyttsx3 voice                                    | Piper TTS, whisper.cpp / Vosk voice commands              |
| Record     | SHA-256 hash-chained log, deterministic trace replay                                       | CFDP downlink package                                     |
| Compute    | Laptop CPU: 13.4 FPS in the slowest 5% of frames                                           | Jetson Orin, within 25 W                                  |

## Working prototype

- Guided web app: choose an experiment, act each step, the engine judges live.
- 14 recorded scenarios replayed through the engine.
- Live webcam mode with printed markers; desktop operator window.
- 602 automated tests; every build gate green.
- Offline by construction: no network path in the on-board code.

`python -m demo`

# Feasibility and viability

## Measured on the working prototype

| Measure                          | Target | CSP-1 · 14 steps | CRX-2 · 8 steps |
|----------------------------------|--------|------------------|-----------------|
| Step accuracy                    | ≥ 95%  | 98.7% met        | 95.6% met       |
| Deviations caught                | ≥ 90%  | 97.8% met        | 97.5% met       |
| False alarms per 45 min          | ≤ 1    | 0.85 met         | 0.00 met        |
| False alarms, permitted reorders | 0      | 0 met            | 0 met           |
| Alert delay, median / 95th pct   | ≤ 2 s  | 1.6 / 1.6 s met  | 1.6 / 1.8 s met |

Degraded synthetic runs — noise, dropped frames, occlusion — each scored against the error injected into it: 130 CSP-1 runs, 70 CRX-2 runs. An alarm counts as false if nothing was injected, or if it concerns a step before the injected one.

## Risks and how we handle them

### Orbit looks different

Learned parts never use gravity cues; geometry is in the rack frame; fine-tune on MicroG-4M.

### False alarms erode trust

Evidence plus persistence; unknown is never wrong; the remaining causes are measured and named.

### The crew's body blocks the view

Reads “cannot verify”; duration priors carry state; multi-camera fusion planned.

### Edge compute and power

Already real-time on a laptop CPU; Jetson Orin with TensorRT INT8 for the 25 W budget.

**Viability** Pure software on standard hardware. It reads the same procedure the crew already follows, is advisory and never commands hardware, and has no network path in the on-board code — a test fails the build if one appears.

# Impact and benefits

SIH 2026 · PS 26174 · Space Technology

## 8 KB

flight record for a 141 s run, against about 70 MB of video

## 0

false alarms on correct, reordered and blocked-camera runs, in both experiments

## 1 file

to add a new experiment — no retraining, no code change

## 13.4 FPS

on a laptop CPU, in the slowest 5% of frames

## Crew and mission

- Fewer experiment runs lost to a missed step.
- Hands-free guidance: attention stays on the task.
- Safety rules catch hazards no checklist can see.

## ISRO operations

- A verifiable record of every run for the principal investigator.
- A tiny downlink footprint that fits constrained links.
- One system across the payload catalogue — procedures are data.

## Society and economy

- Indigenous capability for Gaganyaan and Bharatiya Antariksh Station.
- Transfers to laboratory, pharma and manufacturing SOP compliance.
- Privacy by design: video never has to leave the station.

**Why it has to run on board** A round trip to the Moon takes about 2.6 s and to Mars 6 to 44 minutes. Ground control cannot watch a step as it happens; the witness has to be on the station.

## Datasets and benchmarks

- MicroG-4M — microgravity human activity (arXiv:2506.02845)
- IndustReal — procedure step recognition, WACV 2024 (arXiv:2310.17323)
- Assembly101 — mistake detection, CVPR 2022
- CaptainCook4D — procedural error taxonomy (arXiv:2312.14556)
- EgoPER — error detection in procedural video, CVPR 2024
- PREGO — online mistake detection, CVPR 2024
- HoloAssist — interactive assistance, ICCV 2023

## Methods and tools

- AprilTag 3 — fiducial markers, IROS 2019
- RT-DETR — real-time detection (arXiv:2304.08069)
- MediaPipe Hands; RTMPose / MMPose
- ByteTrack — multi-object tracking (arXiv:2110.06864)
- MS-TCN++ — temporal action segmentation, TPAMI 2020
- Page's CUSUM — sequential change detection
- Piper TTS; whisper.cpp; OpenCV

## Standards and mission context

- NASA-STD-3001 — Space Flight Human-System Standard
- CCSDS 727.0-B — CCSDS File Delivery Protocol (CFDP)
- ISRO — Gaganyaan and Bharatiya Antariksh Station (isro.gov.in)
- Digital Personal Data Protection Act, 2023 (MeitY)
- Vision-based mistake analysis in procedural activities: a review (arXiv:2510.19292)

**Prototype and documentation** run it with `python -m demo` · full write-up in docs/PROJECT\_DOCUMENTATION.md · every figure on these slides is read from runs/ when the deck is built.